

Education Policy Lab

A Longitudinal Analysis of Wage Stagnation in Education Majors

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Acknowledgments



Lab Direction

This project was conducted as part of the Education Policy Lab at Texas State University under the direction of Dr. Li Feng.



Acknowledgments

I thank Dr. Feng, especially for her continued guidance from the very start of this project. I further thank Dr. Mawi, and other members of the EPL (Joe, Skylar, Maria, Monica) for their valuable feedback on the project and presentation.

Motivation; Teacher Shortage in the US

According to Learning Policy Institute – In 2025:

- 31 States + D.C. published vacancies with **45,582** unfilled teaching positions.
- 48 States + D.C. had **365,967** teaching positions filled by teachers who were not fully certified for their teaching assignments.
- Total = **411,549** positions not sufficiently filled
 - This is 1/8th of all teaching positions nationally
- Previous publications show that this has been a problem even since a decade ago.

Known Information and Gaps

Known:

- Evidence of monopsony-like wage dynamics in school districts (Ransom & Sims, 2010).
- Widening wage gaps in different majors (Andrews et. al, 2022)

Gaps:

- Previous research often lacks the 10-year tracking of specific cohorts.
- Degree-specific premium from geographic and state-wise data is lacking.
- Specific flow into industry and geography for education majors, on a national aggregation level, is missing.

Research Questions

- What are the 1-yr, 5-yr, 10-yr post graduation salaries for CS (CIP 11), Education (CIP 13), and Healthcare Professions (CIP 51) majors?
- What are the Industry and Geographic Flows for the three majors?
- To what extent does the regression-adjusted wage premium for Education stagnate relative to STEM and Healthcare Professions over a 10-year horizon?

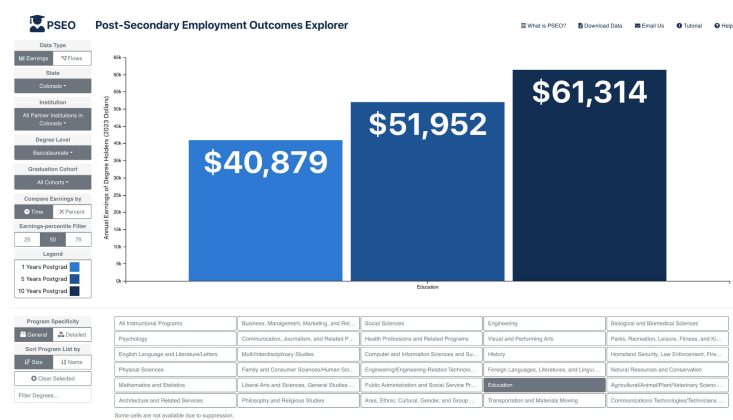
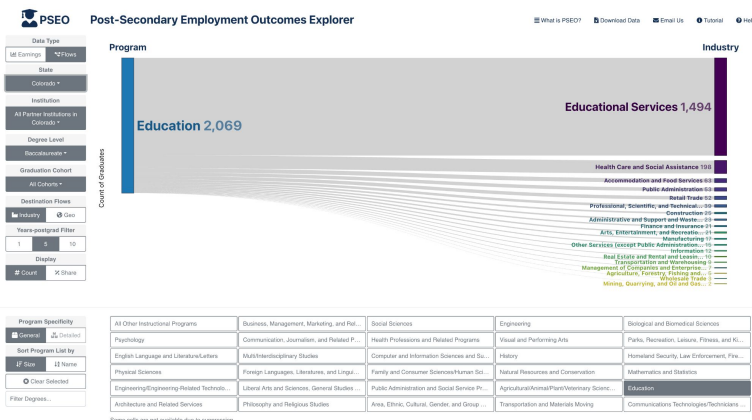
Data: PSEO – US Census Bureau

- Post Secondary Employment Outcomes
- Administrative Records: Links university transcripts to Unemployment Insurance tax records.
- Captures actual earnings for the same cohorts at 1, 5 and 10 years post-grad.
 - 25th, 50th and 75th percentile earnings for each year
- Multiple Aggregation Levels:
 - I used 34 for earnings visualizations and regression (Degree x Major x Cohort x State)
 - 178 for Flow Diagrams (D x M x C x S x Census Division x Industry)
- 2 Digit Zip Code
 - Versatility with the lowest suppression
- Bachelor's Degree Only (Degree = 05)

Dashboard – EPL's vs PSEO Explorer

- You can access EPL's working dashboard at:
<https://edu-wage-stagnation.streamlit.app/>

- PSEO Explorer's Data not Nationally Aggregated

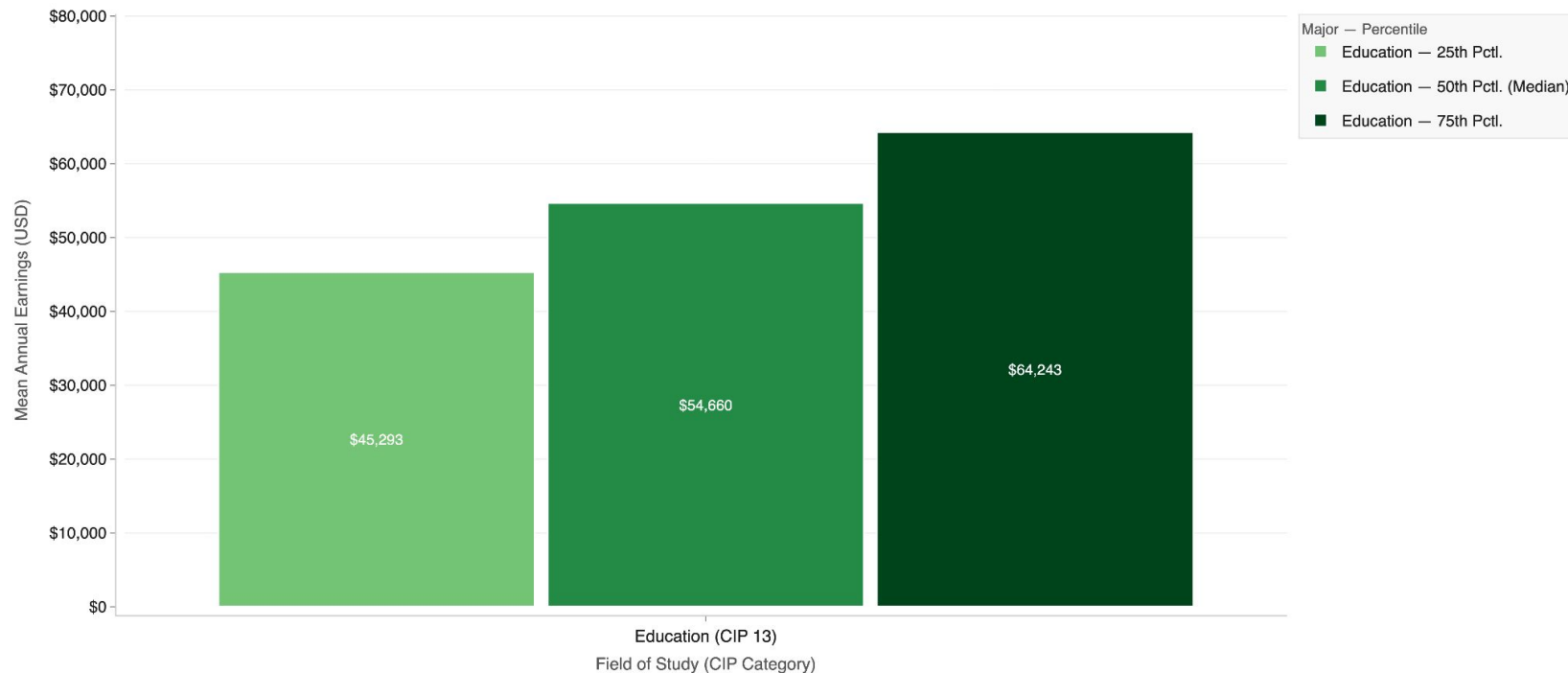


Income By Major

Inflation Adjusted Income – Early Career

5 Years Post-Grad | Education | All States Combined | Cohort: All Cohorts Combined

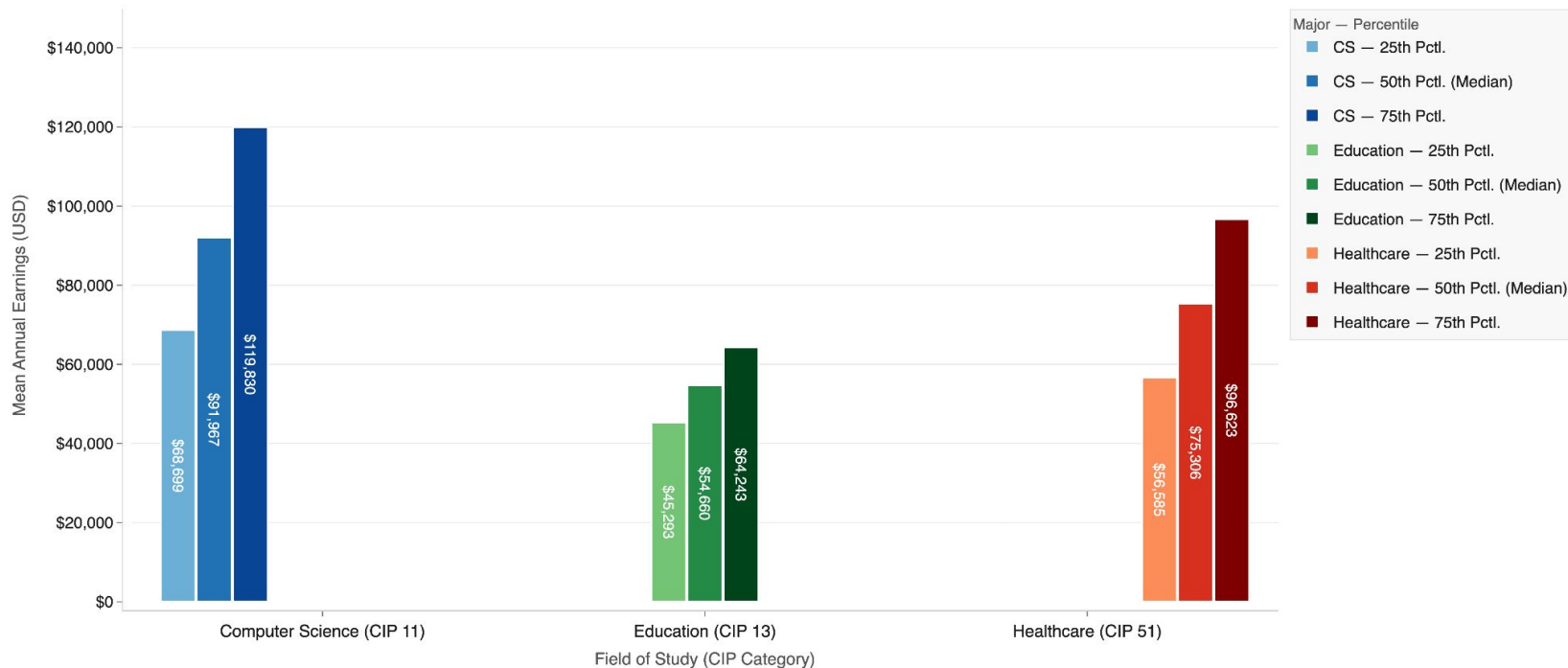
Mean 25th, 50th & 75th percentile earnings · All States Combined · Cohort: All Cohorts Combined



Inflation Adjusted Income – Early Career

5 Years Post-Grad | CS · Education · Healthcare | All States Combined | Cohort: All Cohorts Combined

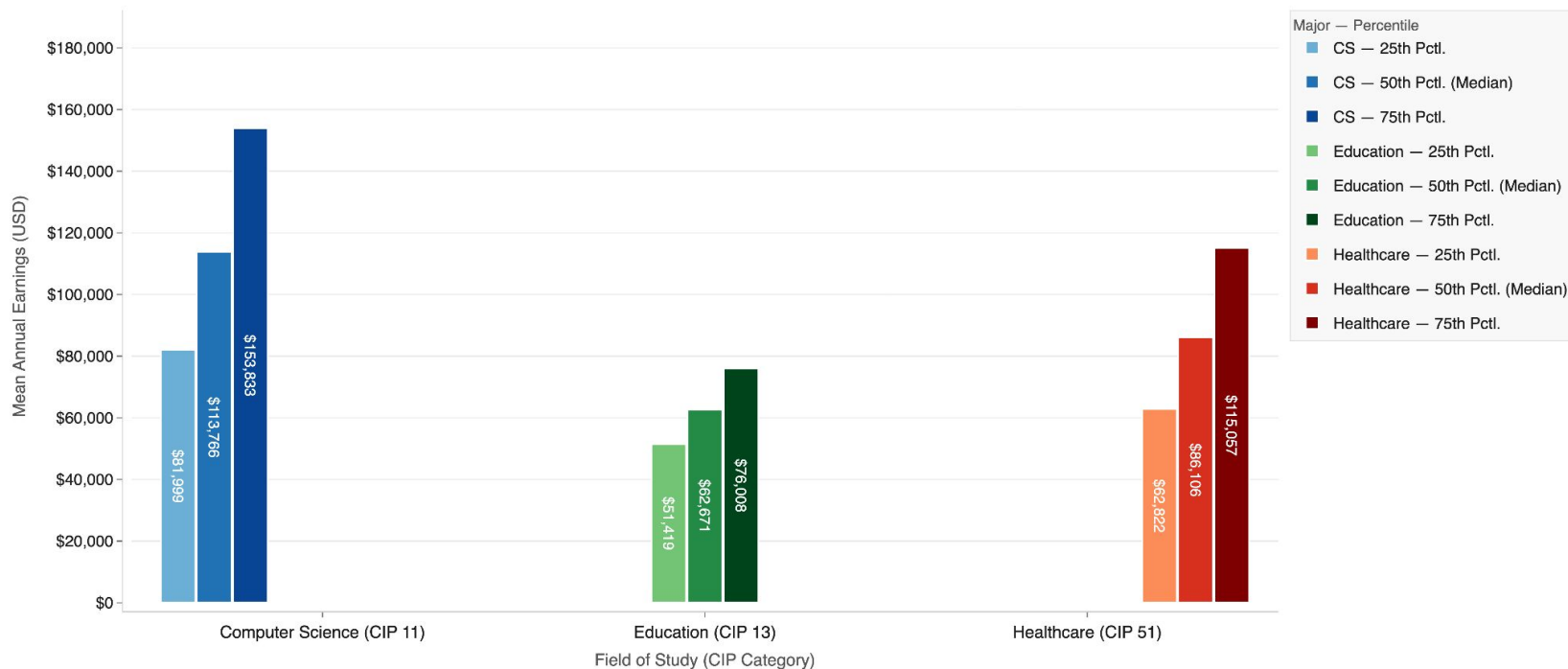
Mean 25th, 50th & 75th percentile earnings · All States Combined · Cohort: All Cohorts Combined



Inflation Adjusted Income – Mid Career

10 Years Post-Grad | CS · Education · Healthcare | All States Combined | Cohort: All Cohorts Combined

Mean 25th, 50th & 75th percentile earnings · All States Combined · Cohort: All Cohorts Combined

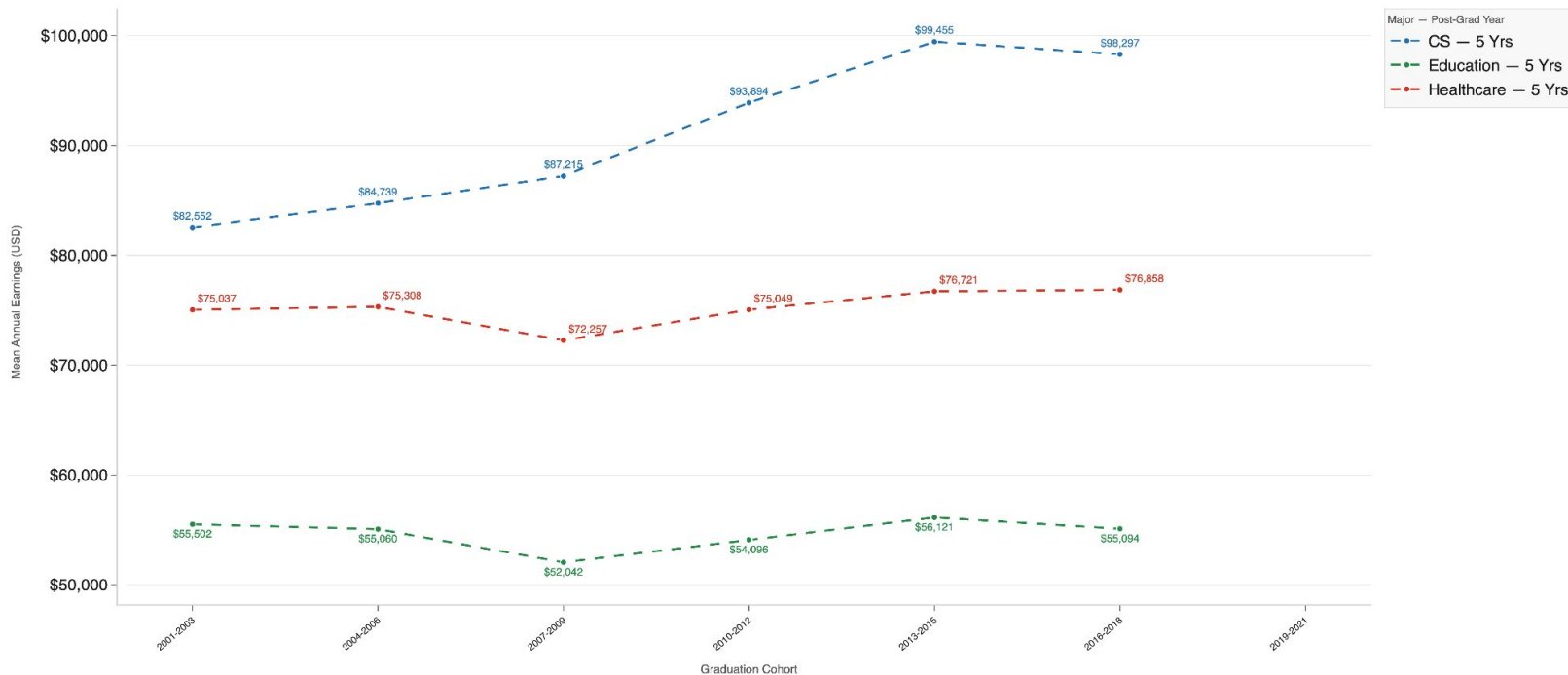


Longitudinal Income View

Earnings Longitudinal View – Early Career – Median

Cohort Earnings — 50th Pctl. (Median) | CS · Education · Healthcare | 5 Yrs Post-Grad | All States Combined

50th Pctl. (Median) mean earnings by graduation cohort · All States Combined · 5 Yrs post-graduation

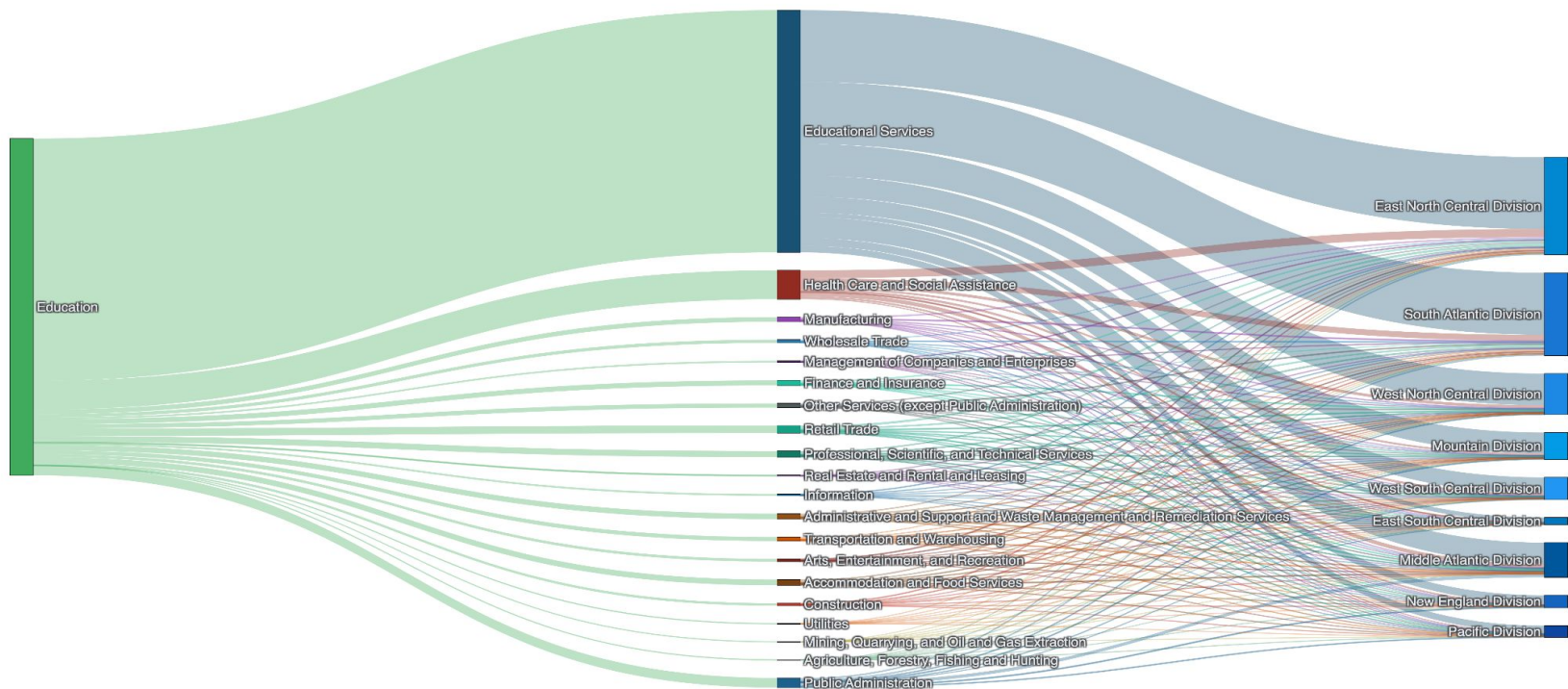


Industry and Geographic Flow

Flow Information – Education – Early Career

5 Years Post-Grad | Education | All States | Cohort: All Cohorts

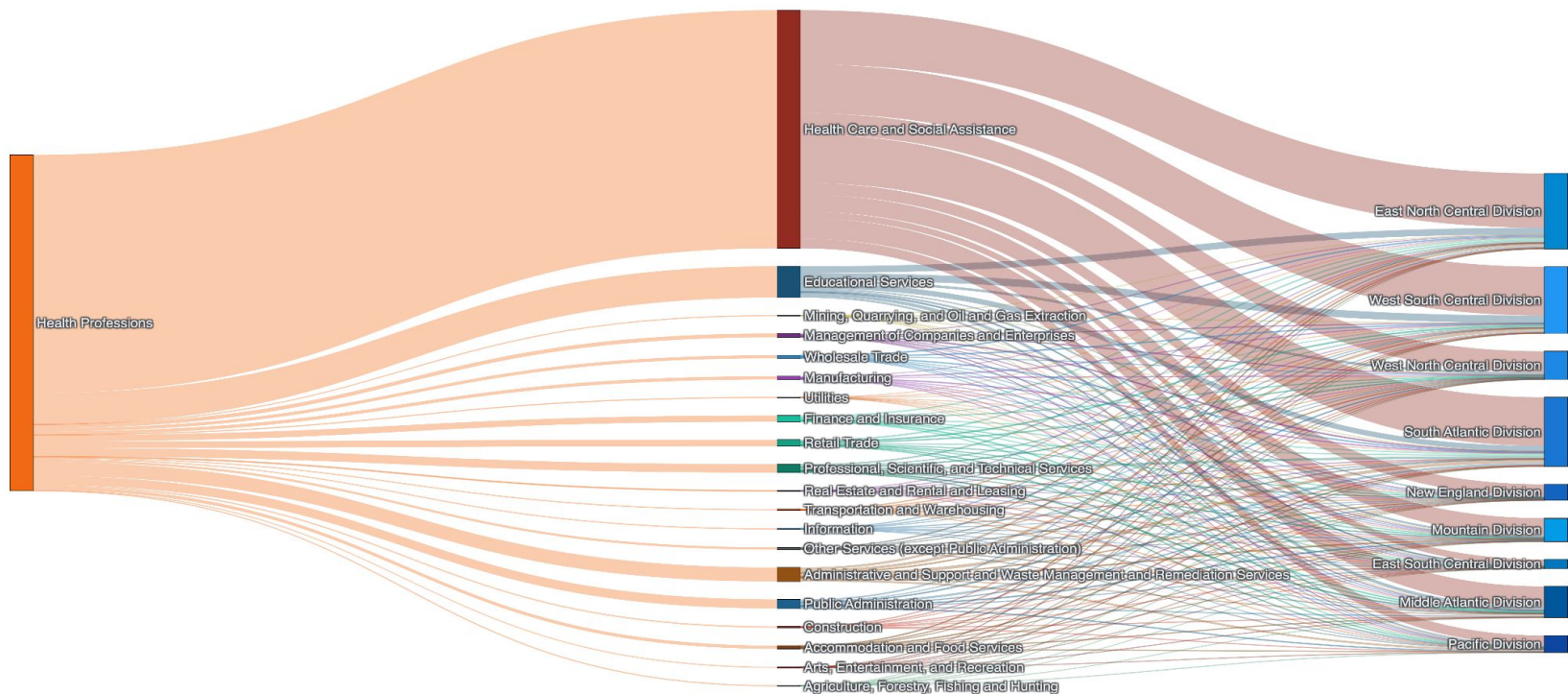
Major → Industry → Geography (In-State / Division) · Agg 178, Degree 05, CIP 11/13/51



Flow Information – Healthcare Professions – Early Career

5 Years Post-Grad | Health Professions | All States | Cohort: All Cohorts

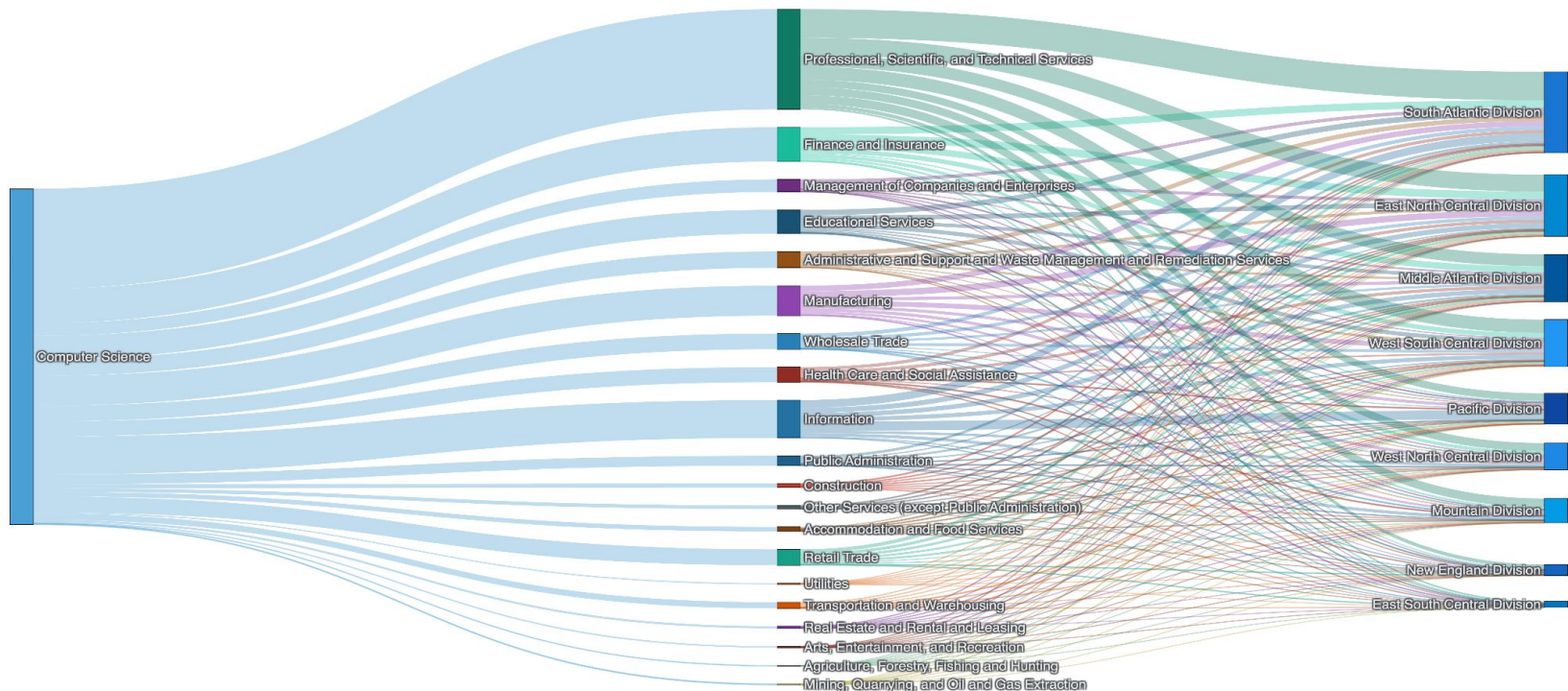
Major → Industry → Geography (In-State / Division) · Agg 178, Degree 05, CIP 11/13/51



Flow Information – CS – Early Career

5 Years Post-Grad | Computer Science | All States | Cohort: All Cohorts

Major → Industry → Geography (In-State / Division) · Agg 178, Degree 05, CIP 11/13/51



Regression Analysis

Regression Models Explained

- Multiple Linear Regression with Dummy (Indicator) Variables.
- The Equation Used:

$$Y_i = \beta_0 + \sum_{j=1}^k \beta_j Major_{i,j} + \sum_{s=1}^m \gamma_s State_{i,s} + \sum_{t=1}^n \delta_t Cohort_{i,t} + \epsilon_i$$

Y_i : Quartile annual earnings ($P_{25/50/75}$) for observation i .

β_0 : Model intercept (Agriculture, 2001 cohort baseline).

β_j : Regression coefficient for major-specific wage premium.

γ_s : Geographic indicators (State-level variation).

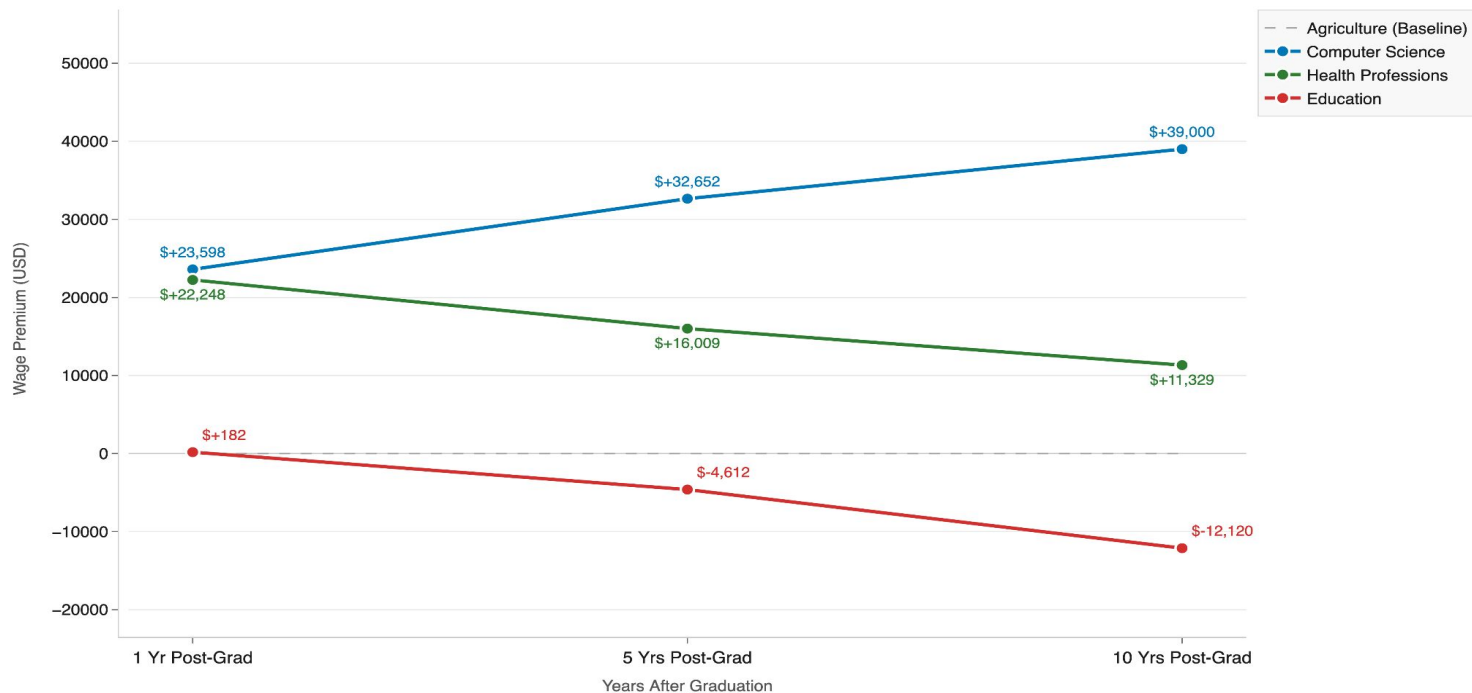
δ_t : Temporal indicators (Macroeconomic cohort trends).

ϵ_i : Stochastic error term (Unobserved heterogeneity).

Wage Premiums Compared to Base CIP 01 – Median

Wage Premium vs. Agriculture (CIP 01) — 50th Percentile

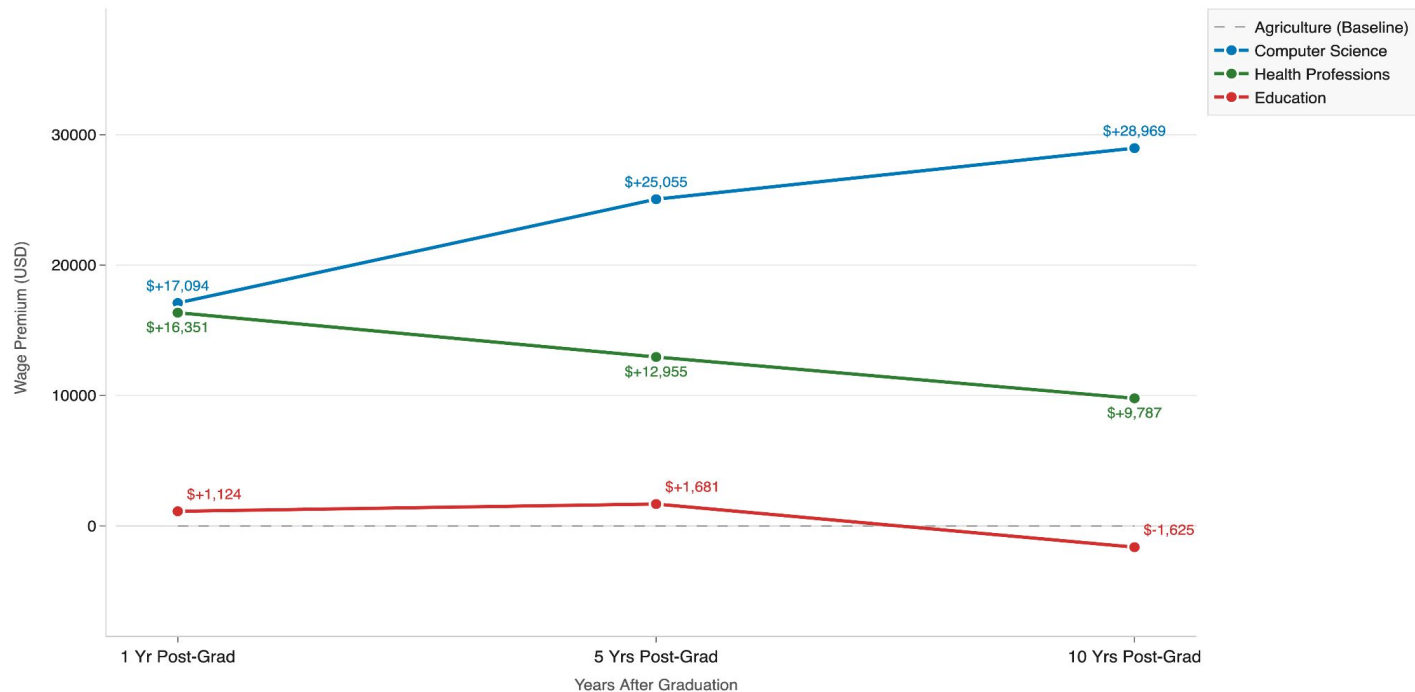
OLS β coefficients · State & cohort fixed effects · 763 observations



Wage Premiums Compared to Base CIP 01 – Lower Quartile

Wage Premium vs. Agriculture (CIP 01) — 25th Percentile

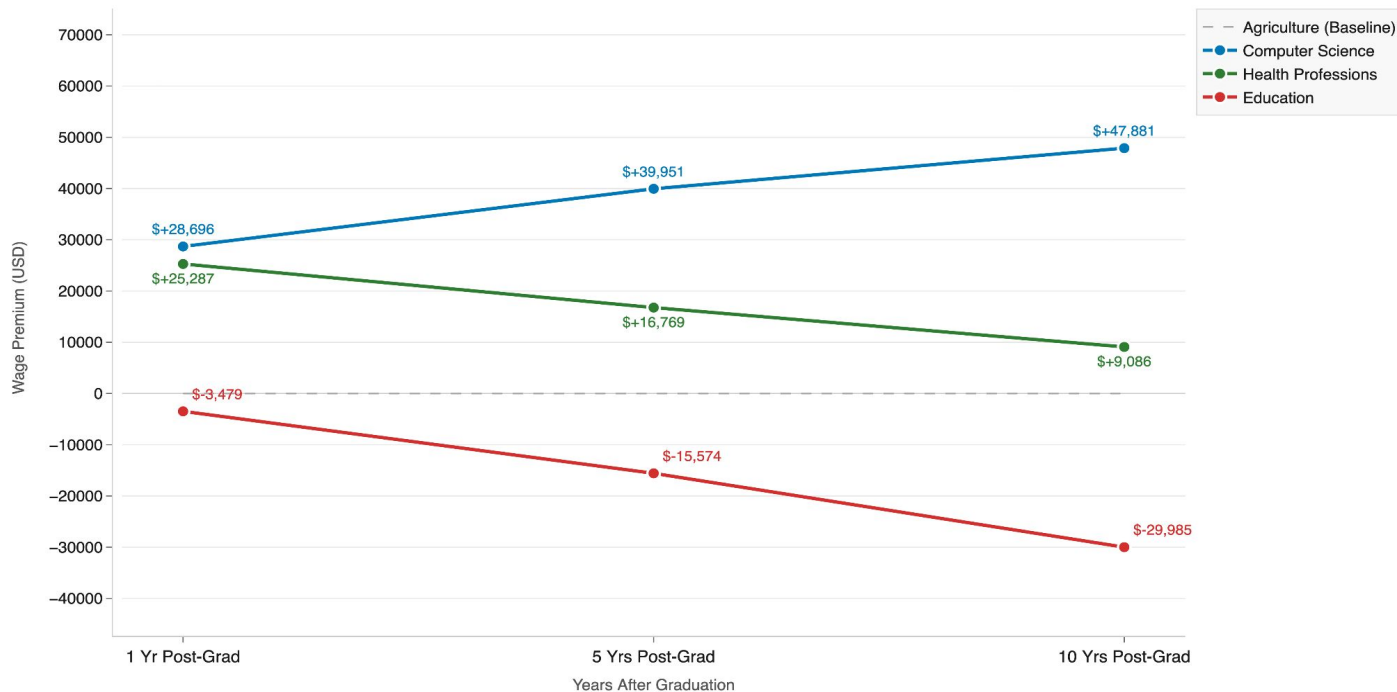
OLS β coefficients · State & cohort fixed effects · 763 observations



Wage Premiums Compared to Base CIP 01 – Upper Quartile

Wage Premium vs. Agriculture (CIP 01) — 75th Percentile

OLS β coefficients · State & cohort fixed effects · 763 observations



Limitations

- PSEO relies on Unemployment Insurance (UI) records; it excludes the self-employed, federal employees, and those working outside participating states.
- Using 2-digit CIP codes (Education) provides data density but loses detail between sub-fields (e.g., Special Ed vs. Physical Ed).
- I am only looking at Bachelor's degree holders, while Master's and further are increasingly more desired for employment.

Future Work

- Integrating Title II Data to analyze specific certifications and influence on wages or some teacher employment data .
- Developing a Geographic Cost-of-Living (COL) Index to further refine the 'State Controls.'
- Analysing the concept of “shortage” further.

References

- *An Overview of Teacher Shortages: 2025*. (2025, July 16). Learning Policy Institute.
<https://learningpolicyinstitute.org/product/overview-teacher-shortages-2025-factsheet>
- Ransom, M., & Sims, David P. (2010). Estimating the Firm's Labor Supply Curve in a "New Monopsony" Framework: Schoolteachers in Missouri. *Journal of Labor Economics*, 28(2), 331–355.
<https://doi.org/10.1086/649904>
- Rodney J. Andrews, Scott A. Imberman, Michael F. Lovenheim, and Kevin M. Stange, "The Returns to College Major Choice: Average and Distributional Effects, Career Trajectories, and Earnings Variability," NBER Working Paper 30331 (2022), <https://doi.org/10.3386/w30331>.
- *PSEO Explorer - Census Bureau*. (2026). Census.gov. <https://lehd.ces.census.gov/applications/pseo/>

Thank You!



Appendix

Regression Tables – Raw

OLS Regression Results

Dep. Variable: y1_p25_earnings **R-squared:** 0.659
Model: OLS **Adj. R-squared:** 0.655
Method: Least Squares **F-statistic:** 161.6
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 2.87e-169
Time: 12:57:26 **Log-Likelihood:** -7705.2
No. Observations: 763 **AIC:** 1.543e+04
Df Residuals: 753 **BIC:** 1.548e+04
Df Model: 9
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.707e+04	763.152	35.465	0.000	2.56e+04	2.86e+04
C(cipcode, Treatment("01"))[T.11]	1.709e+04	614.526	27.817	0.000	1.59e+04	1.83e+04
C(cipcode, Treatment("01"))[T.13]	1123.7695	620.504	1.811	0.071	-94.354	2341.893
C(cipcode, Treatment("01"))[T.51]	1.635e+04	614.543	26.607	0.000	1.51e+04	1.76e+04
C(state_cohort)[T.0_2004]	1624.1895	889.553	1.826	0.068	-122.109	3370.488
C(state_cohort)[T.0_2007]	2495.4835	875.221	2.851	0.004	777.321	4213.646
C(state_cohort)[T.0_2010]	858.7353	839.693	1.023	0.307	-789.682	2507.152
C(state_cohort)[T.0_2013]	2904.3462	836.999	3.470	0.001	1261.217	4547.476
C(state_cohort)[T.0_2016]	3714.6002	836.999	4.438	0.000	2071.471	5357.729
C(state_cohort)[T.0_2019]	3882.5241	866.029	4.483	0.000	2182.406	5582.643
Omnibus:	74.470	Durbin-Watson:	1.833			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	208.700			
Skew:	0.489	Prob(JB):	4.80e-46			
Kurtosis:	5.368	Cond. No.	9.77			

Notes:
 [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

OLS Regression Results

Dep. Variable: y1_p50_earnings **R-squared:** 0.737
Model: OLS **Adj. R-squared:** 0.734
Method: Least Squares **F-statistic:** 234.3
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 1.56e-211
Time: 12:57:26 **Log-Likelihood:** -7827.8
No. Observations: 763 **AIC:** 1.568e+04
Df Residuals: 753 **BIC:** 1.572e+04
Df Model: 9
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	3.854e+04	896.264	42.999	0.000	3.68e+04	4.03e+04
C(cipcode, Treatment("01"))[T.11]	2.36e+04	721.714	32.698	0.000	2.22e+04	2.5e+04
C(cipcode, Treatment("01"))[T.13]	181.5350	728.735	0.249	0.803	-1249.059	1612.129
C(cipcode, Treatment("01"))[T.51]	2.225e+04	721.734	30.826	0.000	2.08e+04	2.37e+04
C(state_cohort)[T.0_2004]	1567.5975	1044.713	1.501	0.134	-483.298	3618.493
C(state_cohort)[T.0_2007]	2560.2251	1027.881	2.491	0.013	542.373	4578.077
C(state_cohort)[T.0_2010]	-99.6437	986.155	-0.101	0.920	-2035.585	1836.297
C(state_cohort)[T.0_2013]	3028.2914	982.992	3.081	0.002	1098.560	4958.022
C(state_cohort)[T.0_2016]	4157.7120	982.992	4.230	0.000	2227.981	6087.443
C(state_cohort)[T.0_2019]	5263.5010	1017.086	5.175	0.000	3266.841	7260.161
Omnibus:	75.827	Durbin-Watson:	1.827			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	265.647			
Skew:	0.423	Prob(JB):	2.07e-58			
Kurtosis:	5.764	Cond. No.	9.77			

Notes:
 [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression Tables – Raw

OLS Regression Results

Dep. Variable: y1_p75_earnings **R-squared:** 0.749
Model: OLS **Adj. R-squared:** 0.746
Method: Least Squares **F-statistic:** 249.2
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 5.22e-219
Time: 12:57:26 **Log-Likelihood:** -7983.6
No. Observations: 763 **AIC:** 1.599e+04
Df Residuals: 753 **BIC:** 1.603e+04
Df Model: 9

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	5.273e+04	1099.169	47.973	0.000	5.06e+04	5.49e+04
C(cipcode, Treatment("01"))[T.11]	2.87e+04	885.102	32.421	0.000	2.7e+04	3.04e+04
C(cipcode, Treatment("01"))[T.13]	-3478.5351	893.713	-3.892	0.000	-5233.000	-1724.070
C(cipcode, Treatment("01"))[T.51]	2.529e+04	885.127	28.568	0.000	2.35e+04	2.7e+04
C(state_cohort)[T.0_2004]	200.8409	1281.224	0.157	0.875	-2314.355	2716.036
C(state_cohort)[T.0_2007]	1263.8238	1260.581	1.003	0.316	-1210.848	3738.496
C(state_cohort)[T.0_2010]	-1537.9280	1209.410	-1.272	0.204	-3912.145	836.289
C(state_cohort)[T.0_2013]	1970.9580	1205.531	1.635	0.102	-395.643	4337.559
C(state_cohort)[T.0_2016]	3522.7041	1205.531	2.922	0.004	1156.103	5889.305
C(state_cohort)[T.0_2019]	4225.7420	1247.342	3.388	0.001	1777.060	6674.424
Omnibus:	244.717	Durbin-Watson:	1.706			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1472.950			
Skew:	1.309	Prob(JB):	0.00			
Kurtosis:	9.283	Cond. No.	9.77			

OLS Regression Results

Dep. Variable: y5_p25_earnings **R-squared:** 0.694
Model: OLS **Adj. R-squared:** 0.690
Method: Least Squares **F-statistic:** 182.8
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 3.61e-160
Time: 12:57:26 **Log-Likelihood:** -6687.9
No. Observations: 653 **AIC:** 1.339e+04
Df Residuals: 644 **BIC:** 1.343e+04
Df Model: 8

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4.157e+04	901.130	46.128	0.000	3.98e+04	4.33e+04
C(cipcode, Treatment("01"))[T.11]	2.505e+04	767.086	32.662	0.000	2.35e+04	2.66e+04
C(cipcode, Treatment("01"))[T.13]	1680.8103	774.761	2.169	0.030	159.447	3202.173
C(cipcode, Treatment("01"))[T.51]	1.295e+04	766.041	16.912	0.000	1.15e+04	1.45e+04
C(state_cohort)[T.0_2004]	953.4991	1026.935	0.928	0.354	-1063.046	2970.044
C(state_cohort)[T.0_2007]	-381.2701	1010.392	-0.377	0.706	-2365.332	1602.791
C(state_cohort)[T.0_2010]	2103.0738	969.382	2.169	0.030	199.542	4006.606
C(state_cohort)[T.0_2013]	4466.6790	966.270	4.623	0.000	2569.259	6364.099
C(state_cohort)[T.0_2016]	3889.8172	970.976	4.006	0.000	1983.156	5796.478
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	166.964	Durbin-Watson:	1.708			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	781.586			
Skew:	1.074	Prob(JB):	1.91e-170			
Kurtosis:	7.910	Cond. No.	inf			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression Tables – Raw

OLS Regression Results

Dep. Variable: y5_p50_earnings **R-squared:** 0.735
Model: OLS **Adj. R-squared:** 0.732
Method: Least Squares **F-statistic:** 223.0
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 5.57e-180
Time: 12:57:26 **Log-Likelihood:** -6870.3
No. Observations: 653 **AIC:** 1.376e+04
Df Residuals: 644 **BIC:** 1.380e+04
Df Model: 8

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	5.664e+04	1191.610	47.529	0.000	5.43e+04	5.9e+04
C(cipcode, Treatment("01"))[T.11]	3.265e+04	1014.357	32.190	0.000	3.07e+04	3.46e+04
C(cipcode, Treatment("01"))[T.13]	-4612.1124	1024.506	-4.502	0.000	-6623.889	-2600.336
C(cipcode, Treatment("01"))[T.51]	1.601e+04	1012.975	15.804	0.000	1.4e+04	1.8e+04
C(state_cohort)[T.0_2004]	402.1705	1357.968	0.296	0.767	-2264.410	3068.751
C(state_cohort)[T.0_2007]	-439.3281	1336.094	-0.329	0.742	-3062.955	2184.298
C(state_cohort)[T.0_2010]	2968.2541	1281.864	2.316	0.021	451.116	5485.392
C(state_cohort)[T.0_2013]	5882.9424	1277.748	4.604	0.000	3373.887	8391.998
C(state_cohort)[T.0_2016]	5236.4759	1283.971	4.078	0.000	2715.200	7757.752
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	374.839	Durbin-Watson:	1.681			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	6220.102			
Skew:	2.198	Prob(JB):	0.00			
Kurtosis:	17.467	Cond. No.	inf			

Notes:

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OLS Regression Results

Dep. Variable: y5_p75_earnings **R-squared:** 0.728
Model: OLS **Adj. R-squared:** 0.725
Method: Least Squares **F-statistic:** 215.8
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 1.32e-176
Time: 12:57:26 **Log-Likelihood:** -7106.0
No. Observations: 653 **AIC:** 1.423e+04
Df Residuals: 644 **BIC:** 1.427e+04
Df Model: 8

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.654e+04	1709.548	44.773	0.000	7.32e+04	7.99e+04
C(cipcode, Treatment("01"))[T.11]	3.995e+04	1455.251	27.453	0.000	3.71e+04	4.28e+04
C(cipcode, Treatment("01"))[T.13]	-1.557e+04	1469.812	-10.596	0.000	-1.85e+04	-1.27e+04
C(cipcode, Treatment("01"))[T.51]	1.677e+04	1453.269	11.539	0.000	1.39e+04	1.96e+04
C(state_cohort)[T.0_2004]	-465.6312	1948.215	-0.239	0.811	-4291.252	3359.989
C(state_cohort)[T.0_2007]	-926.3537	1916.832	-0.483	0.629	-4690.349	2837.642
C(state_cohort)[T.0_2010]	3425.0021	1839.031	1.862	0.063	-186.220	7036.224
C(state_cohort)[T.0_2013]	7227.1945	1833.126	3.943	0.000	3627.568	1.08e+04
C(state_cohort)[T.0_2016]	7970.3061	1842.054	4.327	0.000	4353.148	1.16e+04
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	519.078	Durbin-Watson:	1.777			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	17271.636			
Skew:	3.230	Prob(JB):	0.00			
Kurtosis:	27.353	Cond. No.	inf			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Regression Tables – Raw

OLS Regression Results

Dep. Variable: y10_p25_earnings **R-squared:** 0.697
Model: OLS **Adj. R-squared:** 0.692
Method: Least Squares **F-statistic:** 151.1
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 6.55e-99
Time: 12:57:26 **Log-Likelihood:** -4180.7
No. Observations: 401 **AIC:** 8375.
Df Residuals: 394 **BIC:** 8403.
Df Model: 6

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4.98e+04	1185.974	41.992	0.000	4.75e+04	5.21e+04
C(cipcode, Treatment("01"))[T.11]	2.897e+04	1181.429	24.520	0.000	2.66e+04	3.13e+04
C(cipcode, Treatment("01"))[T.13]	-1625.1300	1189.558	-1.366	0.173	-3963.804	713.544
C(cipcode, Treatment("01"))[T.51]	9787.0013	1178.805	8.302	0.000	7469.466	1.21e+04
C(state_cohort)[T.0_2004]	2905.3084	1236.424	2.350	0.019	474.495	5336.122
C(state_cohort)[T.0_2007]	3722.5013	1216.524	3.060	0.002	1330.810	6114.192
C(state_cohort)[T.0_2010]	5364.7462	1173.057	4.573	0.000	3058.513	7670.979
C(state_cohort)[T.0_2013]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2016]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	161.777	Durbin-Watson:	1.534			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1670.368			
Skew:	1.412	Prob(JB):	0.00			
Kurtosis:	12.592	Cond. No.	inf			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

OLS Regression Results

Dep. Variable: y10_p50_earnings **R-squared:** 0.757
Model: OLS **Adj. R-squared:** 0.754
Method: Least Squares **F-statistic:** 204.9
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 8.16e-118
Time: 12:57:26 **Log-Likelihood:** -4297.6
No. Observations: 401 **AIC:** 8609.
Df Residuals: 394 **BIC:** 8637.
Df Model: 6

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.028e+04	1587.494	44.272	0.000	6.72e+04	7.34e+04
C(cipcode, Treatment("01"))[T.11]	3.9e+04	1581.411	24.661	0.000	3.59e+04	4.21e+04
C(cipcode, Treatment("01"))[T.13]	-1.212e+04	1592.291	-7.612	0.000	-1.53e+04	-8989.617
C(cipcode, Treatment("01"))[T.51]	1.133e+04	1577.898	7.180	0.000	8226.973	1.44e+04
C(state_cohort)[T.0_2004]	3059.9381	1655.024	1.849	0.065	-193.845	6313.721
C(state_cohort)[T.0_2007]	5602.6705	1628.388	3.441	0.001	2401.255	8804.086
C(state_cohort)[T.0_2010]	7880.4949	1570.204	5.019	0.000	4793.470	1.1e+04
C(state_cohort)[T.0_2013]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2016]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	262.396	Durbin-Watson:	1.466			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	5975.228			
Skew:	2.368	Prob(JB):	0.00			
Kurtosis:	21.308	Cond. No.	inf			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Regression Tables – Raw

OLS Regression Results

Dep. Variable: y10_p75_earnings **R-squared:** 0.753
Model: OLS **Adj. R-squared:** 0.750
Method: Least Squares **F-statistic:** 200.6
Date: Tue, 14 Apr 2026 **Prob (F-statistic):** 1.84e-116
Time: 12:57:26 **Log-Likelihood:** -4457.7
No. Observations: 401 **AIC:** 8929.
Df Residuals: 394 **BIC:** 8957.
Df Model: 6
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	9.921e+04	2366.371	41.925	0.000	9.46e+04	1.04e+05
C(cipcode, Treatment("01"))[T.11]	4.788e+04	2357.303	20.312	0.000	4.32e+04	5.25e+04
C(cipcode, Treatment("01"))[T.13]	-2.999e+04	2373.522	-12.633	0.000	-3.47e+04	-2.53e+04
C(cipcode, Treatment("01"))[T.51]	9085.9786	2352.068	3.863	0.000	4461.806	1.37e+04
C(state_cohort)[T.0_2004]	3995.1140	2467.035	1.619	0.106	-855.084	8845.312
C(state_cohort)[T.0_2007]	8739.2468	2427.329	3.600	0.000	3967.110	1.35e+04
C(state_cohort)[T.0_2010]	1.206e+04	2340.598	5.153	0.000	7459.734	1.67e+04
C(state_cohort)[T.0_2013]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2016]	0	0	nan	nan	0	0
C(state_cohort)[T.0_2019]	0	0	nan	nan	0	0
Omnibus:	342.210	Durbin-Watson:	1.568			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	14763.100			
Skew:	3.274	Prob(JB):	0.00			
Kurtosis:	31.995	Cond. No.	inf			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.